Digital Material Passport

ID **DMP-2024-ARR-001** - *Version* **0.1.0**
Issue Date **2024-12-15** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
Advanced Steel Technologies GmbH Industriestraße 15 52070 Aachen DE quality@advanced-steel.de	**Precision Engineering Ltd.** Manufacturing Park 42 Birmingham B1 1AA GB procurement@precision-eng.co.uk	*Order ID* **PO-2024-7890** *Delivery ID* **DN-2024-3456** - Quantity **500 kg**

Product

Product Name	Hardenability Test Steel	*Shape*	RoundBar - D 100 mm - Length 1000 mm
Batch ID	HTB-2024-045		

Chemical Analysis

Heat Number **H-2024-078** - *Sample Location* **Ladle**

Symbol	*Unit*	*Min*	*Max*	*Actual*
C	%	0.42	0.5	0.45
Mn	%	0.5	0.8	0.65
Si	%		0.4	0.25
P	%		0.03	0.015

Symbol	*Unit*	*Min*	*Max*	*Actual*
S	%		0.025	0.008
Cr	%		0.25	0.12
Ni	%		0.25	0.08

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Jominy Hardenability Test Quenched from 850°C					ISO 377	✓

Distance from - quenched end (mm)	**1.5**	**3**	**5**	**7**	**9**	**11**	**15**	**20**	**25**	**30**
Value [HRC]	45	43	40	36	33	31	27	25	28	27
Min	42	39	35	32	29	26	22	20		
Max	47	46	44	41	39	37	33	31	<= 30	<= 29
Status	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Coating Thickness Profile Magnetic induction method					ISO 2178	✓

Position along length (mm)	**0**	**50**	**100**
Value [µm]	125	115	108
Min	>= 120	>= 110	>= 100
Max	<= 130	<= 125	<= 120
Status	Pass	Pass	Pass

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer, Quality Control Laboratory - 2024-12-15